

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 2.1: INTRODUCTION TO SALTS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Use everything you learned in Lessons 1 to 8 to answer the driving question: Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same, and what does understanding salts tell us about how to prepare and use them safely in Kenya? Write your answers in the space under each prompt. Use evidence from the experiments you did, such as the solubility test, the preparation of salts and the watch-glass experiment. Write chemical formulae and balanced equations with state symbols where asked. Write in full sentences and use the scientific words you learned. Link each answer back to the storeroom and laboratory phenomenon.

Part 1: What is a salt, and why does classification not explain the puzzle?

Remember the three substances from Lesson 1: CAN fertiliser (calcium ammonium nitrate, used on maize farms in the Rift Valley), washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) and table salt (NaCl). (a) Define a salt. (b) Name the four types of salts and give one example of each. (c) The fertiliser, washing soda and table salt are all normal salts. Explain why this fact alone cannot explain why they behaved differently overnight.

Write your answer here:

Part 2: Solubility and why liquid water is not the whole answer

In Lesson 2 you added seven salts to water in test tubes. (a) Which of the salts you tested dissolved, and which did not? Include the salts from the phenomenon. (b) State the solubility rules you used for nitrates, chlorides, carbonates and sulphates. (c) The fertiliser in the storeroom got wet even though no liquid water was added to it. Explain why knowing that a salt is soluble in liquid water is not enough to explain the clumping, and what else must be involved.

Write your answer here:

Part 3: How salts are prepared and why the method depends on solubility

(a) Write a balanced equation with state symbols for preparing zinc sulphate from zinc and dilute sulphuric acid, and say how you identified the gas produced. (b) Write a balanced equation with state symbols for preparing calcium chloride from hydrochloric acid and calcium carbonate, and say how you identified the gas. (c) Which method would you use to prepare barium sulphate, and why can you not use the methods in (a) and (b)? Write the net ionic equation. (d) Why do we add the solid in excess when preparing a soluble salt, and how is the excess removed?

Write your answer here:

Part 4: Explaining the phenomenon using behaviour of salts in air

In Lesson 6 you left calcium chloride, sodium carbonate decahydrate and sodium chloride on watch glasses in classroom air and recorded changes in mass. (a) Describe what happened to each of the three samples and what this showed about its mass. (b) Define deliquescent, efflorescent and stable (unaffected) salts. (c) Use these ideas to explain, for each substance, what you saw in the storeroom and laboratory: the CAN fertiliser, the washing soda and the table salt. Write the equation for what happens to washing soda.

Write your answer here:

Part 5: Using salts safely in Kenya

Use your explanation from Part 4 to answer the second half of the driving question. (a) Give advice to a Rift Valley maize farmer on how to store CAN fertiliser and explain why, in terms of what the salt does in air. (b) Explain how the property of washing soda affects how it must be stored, and why it is still useful in laundry. (c) Choose one other Kenyan use of a salt from Lesson 7 (for example fish preservation at Dunga Beach, ORS, glass making or paint) and explain how the property of the salt makes it suitable. (d) Explain why choosing the right preparation method matters when making salts for use.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of salt definitions, classification and solubility rules	All definitions, the four salt types with correct examples and the solubility rules are correct and complete.	Most definitions, types and rules are correct, with minor omissions or one wrong example.	Definitions or rules are missing, incorrect or confused.
Preparation methods and chemical equations	Balanced equations with correct state symbols, correct gas tests, a correct net ionic equation, and a clear reason why the method depends on solubility.	Equations are mostly correct but have minor errors in balancing or state symbols, or the reason for the method is only partly explained.	Equations are incorrect or missing, or methods are mixed up.
Explanation of the phenomenon (air behaviour)	Correctly explains all three substances using deliquescent, efflorescent and stable behaviour, with evidence from the watch-glass experiment and the washing soda equation.	Explains two or three substances correctly but gives limited evidence or reasoning.	Gives descriptions only, or uses the terms incorrectly.
Link to safe use in Kenya	Gives specific, accurate storage advice and links at least one other Kenyan application to the property of the salt.	Gives correct advice but the link to properties is general.	Advice is vague or not linked to the properties of salts.
Use of evidence and scientific communication	Cites evidence from the lessons, uses scientific vocabulary correctly and writes clearly in full sentences.	Uses some evidence and mostly correct vocabulary.	Little evidence is used and the vocabulary is often incorrect.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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